

The new **Green Series Controller**



Keep temperatures under
control at all times.

ONE CONTROLLER FOR ALL APPLICATIONS



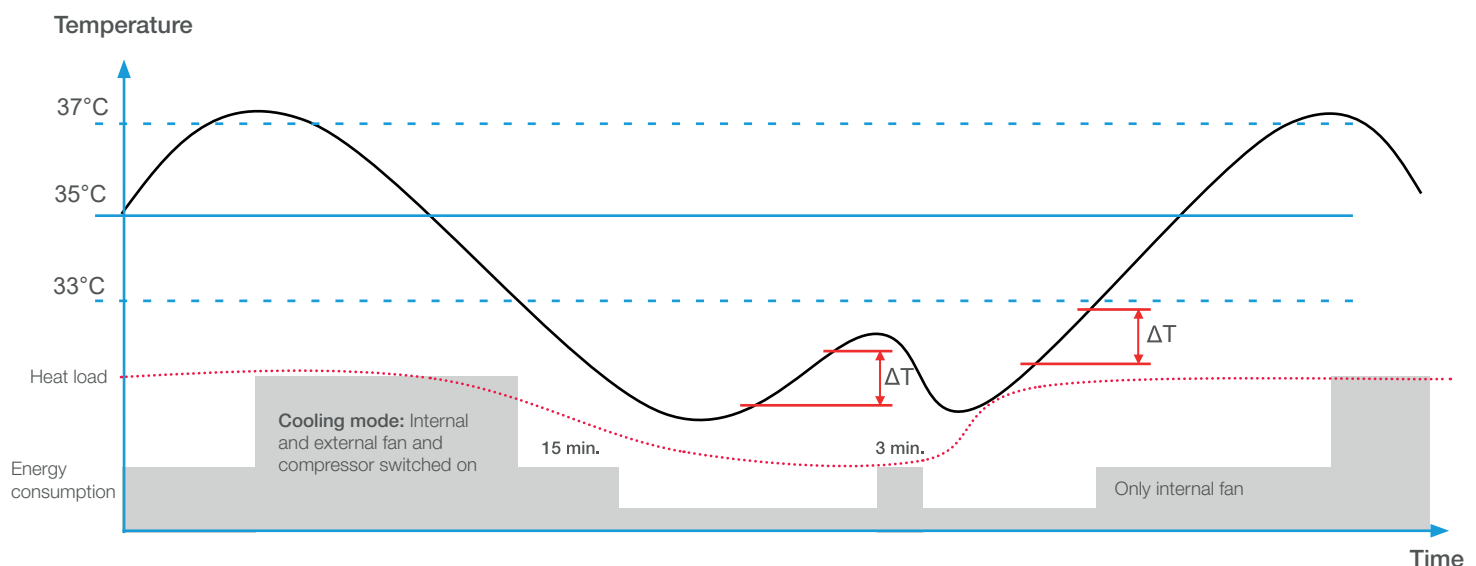
The new Green Series Controller combines all the important functions that were previously separated in the Standard Controller (SC) and Multicontroller (MC). The proven two-point control ensures that the temperature in the control cabinet is reliably maintained. Supplemented by the now even more efficient energy-saving mode in all variants, energy is consistently saved.

Advantages of the new Green Series controller

- A door contact switches the cooling unit off when the enclosure door is open.
- A collective fault message enables the output of any alarm type to a service interface or the output of a simple visual alarm. Supported alarm types are: max. temperature | pressure in the coolant circuit too high | incorrect device setting on the controller | temperature sensor defective.
- An integrated self-test, which can be performed manually at any time, enables OEMs to perform a simple and quick function test and to recognise the customer's machine availability.
- A service interface on the rear of the cooling unit, in conjunction with the 'Pfannenberg Control Centre' software, enables the temperature and error codes to be read out and recorded, and the set temperature and switching hysteresis to be set. A USB converter is required, which is available as an accessory.
- The energy-saving mode is now integrated in every controller and is designed to be even more efficient.

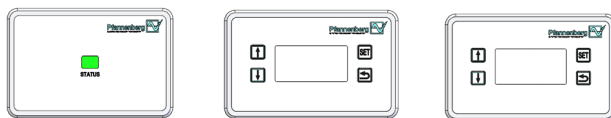
The energy-saving mode

In energy-saving mode, the internal fan is switched off after 15 minutes without a cooling request. Maximum energy is saved. If the controller detects an increase in temperature, it switches the internal fan on for three minutes and circulates the air. If the setpoint temperature is not exceeded, the internal fan switches off again. By detecting a temperature rise regardless of the setpoint temperature, the energy-saving mode can be extended to the maximum, thus saving the most energy. The additional temperature sensor inside the housing can be dispensed with.



The controller is available in three versions

The LED variant (LED) has an operating display that can only indicate fault-free operation or an error. The setpoint temperature is set via DIP switches on the controller board. The display variant (DIS) has a control element with which the parameters of the cooling unit can be changed externally. This variant also has a display for the actual temperature and possible error codes. The Connect variant (CON) has a control element like the Display variant. In addition, this variant is equipped with an RS485 interface that works with the Modbus RTU protocol.



	LED	DIS	CON
Condensate Management	X (not for DTFS)	X	X
Status LED	X	-	-
Main fault contact	X	X	X
Door contact	X	X	X
Integrated self test	-	X	X
Service interface	X	X	X
Pfannenberg Control Center	X	X	X
Antifreeze control	optional	optional	optional
Energy-saving mode	X	X	X
Temperature display	-	X	X
Set buttons	-	X	X
Modbus interface	-	-	X
Family Mode	-	-	X
DIP switch	X	X	X
Blackbox data storage	X	X	X

Condensate Management

Occurring condensate water will be collected in a tray and with using the power of the hot gas pipe evaporated.

Antifreeze Control

A separate temperature sensor prevents the internal heat exchanger from icing up.

Dip switch

Small dip switches are directly on the controller board to adjust the set temperature

Blackbox data storage

In case of power supply loss, major values from the last minutes are stored. You'll get access with the Pfannenberg control center.

New interface and family mode

New options are the Modbus RTU interface and the family mode. As a Connect variant (CON), which is only available with an operating element, the service interface has been replaced by an RS 485 interface with Modbus RTU protocol. This enables particularly easy integration into machine communication. Set and actual values as well as error codes can be conveniently read out via the machine control system. The same interface can also be used to interconnect several devices that are installed on a control cabinet and operate in a network. In family mode, the cooling task is distributed to equalise operating times. If one appliance is in operation more frequently than other cooling appliances, cooling appliances with fewer operating hours are preferentially switched on. This means that all cooling units are loaded equally, which increases the service life and reliability.

With Family Mode, you can protect your cooling appliances by switching them off at high operating hours and homogeneous heat distribution. This ensures fewer cycles, more consistent cooling behaviour and a longer service life for your appliances.

Interested in finding out more about our Green Series controllers?

Please get in touch with your Pfannenberg contact person or send us a [message](#)!